ELEC2302/9302 Signals and Systems

Bonus quiz — version A

Name: _____ unikey: _____

Total marks 18.

You are eligible for this bonus quiz if your highest Quiz 2 mark so far is $\geq 85\%$ (17/20). If you do not meet this requirement you will not receive bonus marks.

Show all working. Use the large boxes for **working** and the small labelled boxes for **final answers**. The formula sheets are attached at the back.

1. Let $u(t) = e^{-t}H(t)$. **[9 marks]**

   (a) Compute the energy $E(u)$. **[1 mark]**

   $E(u) =$

   (b) Find $U(j\omega)$ and use Parseval's theorem to evaluate

   $$\int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{1+\omega^2}.$$ **[2 marks]**

   $U(j\omega) =$

   Value of the integral =

(c) The signal u is fed through an RC lowpass filter with impulse response $h(t) = e^{-t}H(t)$. Find the output $y(t)$. Use Parseval's theorem and the identity

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{(1+\omega^2)^2} = \frac{\pi}{2}$$

to compute $E(y)$. **[2 marks]**

$y(t) =$

$E(y) =$

(d) Let $v(t)$ have Fourier transform $V(j\omega) = 1$ for $|\omega| < W$ and $V(j\omega) = 0$ otherwise, where $W > 0$. Find $v(t)$. **[2 marks]**

$v(t) =$

(e) The signal v passes through an ideal lowpass filter with frequency response

$$G(j\omega) = \begin{cases} 1, & |\omega| < \omega_c, \\ 0, & |\omega| \geq \omega_c, \end{cases}$$

where $0 < \omega_c < W$. What fraction of the energy of v reaches the output? **[2 marks]**

Fraction of the energy of v at the output =

2. Let $r(t) = t\,H(t)$ be the unit ramp, with Fourier transform $R(j\omega)$. **[6 marks]**

(a) Show that

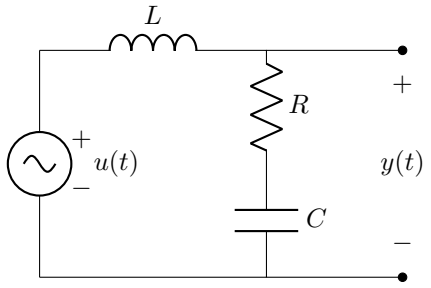
$$H(t) = \int_{-\infty}^{t} \delta(\varsigma)\,\mathrm{d}\varsigma \qquad \text{and} \qquad r(t) = \int_{-\infty}^{t} H(\varsigma)\,\mathrm{d}\varsigma.$$

Use the differentiation property to show that

$$(j\omega)^2 R(j\omega) = 1. \qquad \textbf{[2 marks]}$$

(b) In the circuit below, the frequency response from u to y is

$$G(j\omega) = \frac{1 + 2j\omega}{(1 + j\omega)^2}.$$



The input is the ramp, $u(t) = r(t)$. Define $e(t) = u(t) - y(t)$. Use part (a) to find $E(j\omega)$, and hence find $e(t)$ and $y(t)$. What happens to $e(t)$ as $t \to \infty$? **[4 marks]**

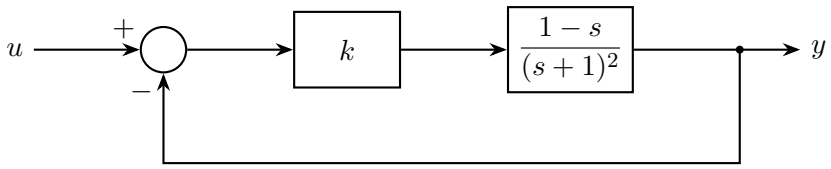
$E(j\omega) =$

$e(t) =$

$y(t) =$

As $t \to \infty$, $e(t) \to$

3. The block diagram below shows the system $G(s) = \dfrac{1-s}{(s+1)^2}$ in a negative feedback loop with a gain $k > 0$. For what range of $k > 0$ is the closed loop system from u to y exponentially stable? **[3 marks]**



Range of k:

ELEC2302/9302 Signals and Systems — Quiz 2 formula sheet

Fourier series (u T-periodic)

Exponential form:

$$\phi_k(t) = e^{jk\omega_0 t}, \qquad c_k = \langle u, \phi_k \rangle_P = \frac{1}{T}\int_0^T u(t)\, e^{-jk\omega_0 t}\,\mathrm{d}t, \qquad u(t) = \sum_{k=-\infty}^{\infty} c_k\, e^{jk\omega_0 t}, \qquad c_0 = \frac{1}{T}\int_0^T u(t)\,\mathrm{d}t.$$

Sine–cosine form (u real): $u(t) = a_0 + \sum_{k=1}^{\infty}\big(a_k\cos(k\omega_0 t) + b_k \sin(k\omega_0 t)\big)$, where

$$a_0 = c_0, \quad a_k = 2\,\mathrm{Re}(c_k) = \frac{2}{T}\int_0^T u(t)\cos(k\omega_0 t)\,\mathrm{d}t, \quad b_k = -2\,\mathrm{Im}(c_k) = \frac{2}{T}\int_0^T u(t)\sin(k\omega_0 t)\,\mathrm{d}t, \quad c_k = \tfrac{1}{2}\,(a_k - jb_k)\,.$$

u real	$c_{-k} = \overline{c_k}$	linearity	$\alpha u_1 + \beta u_2$ has coefficients $\alpha c_{1,k} + \beta c_{2,k}$
u even	$b_k = 0$	time shift	$u(t - t_0)$ has coefficients $e^{-jk\omega_0 t_0}\, c_k$
u odd	$a_0 = a_k = 0$	Parseval	$P(u) = \frac{1}{T}\int_0^T \lvert u(t)\rvert^2\,\mathrm{d}t = \sum_{k=-\infty}^{\infty} \lvert c_k\rvert^2$

Fourier series of standard waveforms

$\mathrm{sinc}(x) = \dfrac{\sin(\pi x)}{\pi x}$.

Signal	Coefficients
$\cos(\omega_0 t)$	$c_1 = c_{-1} = \frac{1}{2}$, all others zero
$\sin(\omega_0 t)$	$c_1 = -\frac{j}{2}$, $\quad c_{-1} = \frac{j}{2}$, all others zero
Square wave, even: $+A$ for $\lvert t\rvert < T/4$, $-A$ for $T/4 < \lvert t\rvert < T/2$	$c_k = A\,\mathrm{sinc}\left(\frac{k}{2}\right)$ (k odd), $\quad c_k = 0$ (k even)
Pulse train: $A\,p_\tau(t)$ repeated with period T, duty cycle $d = \tau/T$	$c_k = A\,d\,\mathrm{sinc}(kd)$
Triangle wave: $A\,(1 - 4\lvert t\rvert/T)$ for $\lvert t\rvert \le T/2$	$c_k = \dfrac{4A}{\pi^2 k^2}$ (k odd), $\quad c_k = 0$ (k even)
Rectified sine: $A\,\lvert\sin(\omega_m t)\rvert$, period $T = \pi/\omega_m$, so $\omega_0 = 2\omega_m$	$c_k = \dfrac{2A}{\pi\,(1 - 4k^2)}$

Fourier transform

$$U(j\omega) = \int_{-\infty}^{\infty} u(t)\, e^{-j\omega t}\,\mathrm{d}t, \qquad u(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} U(j\omega)\, e^{j\omega t}\,\mathrm{d}\omega, \qquad u \text{ periodic: } U(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} c_k\,\delta(\omega - k\omega_0).$$

$u(t)$	$U(j\omega)$	$u(t)$	$U(j\omega)$
$\delta(t)$	1	$p_\tau(t)$	$\tau\,\mathrm{sinc}\left(\frac{\omega\tau}{2\pi}\right)$
$\delta(t - t_0)$	$e^{-j\omega t_0}$		
1	$2\pi\delta(\omega)$	$1 - \dfrac{\lvert t\rvert}{\tau}$ for $\lvert t\rvert < \tau$	$\tau\,\mathrm{sinc}^2\left(\frac{\omega\tau}{2\pi}\right)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$	$e^{-at}H(t)$, $a > 0$	$\dfrac{1}{a + j\omega}$
$\cos(\omega_0 t)$	$\pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$	$t\,e^{-at}H(t)$	$\dfrac{1}{(a + j\omega)^2}$
$\sin(\omega_0 t)$	$-j\pi\delta(\omega - \omega_0) + j\pi\delta(\omega + \omega_0)$	$e^{-a\lvert t\rvert}$	$\dfrac{2a}{a^2 + \omega^2}$
$H(t)$	$\pi\delta(\omega) + \dfrac{1}{j\omega}$	$e^{-at}\cos(\omega_0 t)H(t)$	$\dfrac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$
$\mathrm{sgn}(t)$	$\dfrac{2}{j\omega}$	$e^{-at}\sin(\omega_0 t)H(t)$	$\dfrac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$

Properties of the Fourier transform

	$u(t)$	$U(j\omega)$		$u(t)$	$U(j\omega)$
linearity	$\alpha u_1(t) + \beta u_2(t)$	$\alpha U_1(j\omega) + \beta U_2(j\omega)$	differentiation	$\dfrac{\mathrm{d}}{\mathrm{d}t}u(t)$	$j\omega\, U(j\omega)$
time shift	$u(t - t_0)$	$e^{-j\omega t_0}\, U(j\omega)$	integration	$\displaystyle\int_{-\infty}^{t} u(\varsigma)\,\mathrm{d}\varsigma$	$\dfrac{U(j\omega)}{j\omega} + \pi U(0)\delta(\omega)$
frequency shift	$e^{j\omega_0 t}u(t)$	$U\big(j(\omega - \omega_0)\big)$	multiply by t	$t\,u(t)$	$j\dfrac{\mathrm{d}}{\mathrm{d}\omega}U(j\omega)$
modulation	$u(t)\cos(\omega_0 t)$	$\frac{1}{2}U\big(j(\omega - \omega_0)\big) + \frac{1}{2}U\big(j(\omega + \omega_0)\big)$	convolution	$(h * u)(t)$	$H(j\omega)\, U(j\omega)$
time scaling	$u(at)$	$\dfrac{1}{\lvert a\rvert}\, U\left(\dfrac{j\omega}{a}\right)$	multiplication	$u(t)\, y(t)$	$\dfrac{1}{2\pi}\,(U * Y)(j\omega)$
time reversal	$u(-t)$	$U(-j\omega)$	duality	$U(jt)$	$2\pi\, u(-\omega)$

Parseval's theorem:

$$\int_{-\infty}^{\infty} u(t)\,\overline{y}(t)\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} U(j\omega)\,\overline{Y}(j\omega)\,\mathrm{d}\omega, \qquad E(u) = \int_{-\infty}^{\infty} |u(t)|^2\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |U(j\omega)|^2\,\mathrm{d}\omega.$$

Second order systems

$$G(j\omega) = \frac{K\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}, \qquad G(0) = K,$$

where $\omega_n > 0$ is the natural frequency and ζ the damping ratio.

$\zeta > 1$	overdamped	distinct real poles
$\zeta = 1$	critically damped	repeated pole $-\omega_n$
$\zeta < 1$	underdamped	$-\zeta\omega_n \pm j\omega_d$, $\omega_d = \omega_n\sqrt{1-\zeta^2}$ the damped natural frequency

$$\omega_r = \omega_n\sqrt{1 - 2\zeta^2}, \qquad |G(j\omega_r)| = \frac{|K|}{2\zeta\sqrt{1-\zeta^2}}, \qquad \zeta < \frac{1}{\sqrt{2}}.$$

Stability

Exponentially stable: there exist $M > 0$ and $a > 0$ such that $|h(t)| \le Me^{-at}$ for all $t \ge 0$.
Marginally stable: there exists $M > 0$ such that $|h(t)| \le M$ for all $t \ge 0$, and $h(t) \not\to 0$ as $t \to \infty$.
BIBO stable: $\sup_{t\ge 0}|u(t)| < \infty$ implies $\sup_{t\ge 0}|y(t)| < \infty$ for every input u.

ELEC2302/9302 Signals and Systems — Quiz 1 formula sheet

Complex exponentials

$$e^{j\theta} = \cos\theta + j\sin\theta$$

A complex number in Cartesian and polar form:

$$z = x + jy = r\,e^{j\theta}, \qquad r = |z| = \sqrt{x^2+y^2}, \qquad \theta = \arg z.$$

Inner product, energy and power

$$\langle u, y\rangle := \int_{-\infty}^{\infty} u(t)\,\overline{y}(t)\,\mathrm{d}t$$

$$E(u) = \int_{-\infty}^{\infty} |u(t)|^2\,\mathrm{d}t, \qquad P(u) = \lim_{W\to\infty} \frac{1}{2W}\int_{-W}^{W} |u(t)|^2\,\mathrm{d}t, \qquad u_{\mathrm{RMS}} = \sqrt{P(u)}.$$

If u is periodic with period T, then

$$\langle u, y\rangle_P = \frac{1}{T}\int_0^T u(t)\,\overline{y}(t)\,\mathrm{d}t, \qquad P(u) = \langle u, u\rangle_P = \frac{1}{T}\int_0^T |u(t)|^2\,\mathrm{d}t.$$

Standard signals

Unit step (Heaviside).

$$H(t) = \int_{-\infty}^{t} \delta(\varsigma)\,\mathrm{d}\varsigma = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

Pulse of width T.

$$p_T(t) = \begin{cases} 1, & |t| \leq T/2 \\ 0, & \text{otherwise} \end{cases}$$

$$p_T(t) = H\left(t + \tfrac{T}{2}\right) - H\left(t - \tfrac{T}{2}\right), \qquad \frac{\mathrm{d}}{\mathrm{d}t}H(t) = \delta(t).$$

The Dirac delta

Sifting property.

$$\int_{-\infty}^{\infty} \delta(t-\tau)\,y(t)\,\mathrm{d}t = y(\tau).$$

Scaling.

$$\delta(\alpha t) = \frac{1}{|\alpha|}\,\delta(t).$$

Convolution

$$(h * u)(t) := \int_{-\infty}^{\infty} h(t-\tau)\,u(\tau)\,\mathrm{d}\tau = (u * h)(t)$$

$$(u * \delta)(t) = u(t), \qquad \big(u * \delta(t-T)\big) = u(t-T), \qquad h_1 * (h_2 * h_3) = (h_1 * h_2) * h_3.$$

Impedances

$$Z_R = R, \qquad Z_L = j\omega L, \qquad Z_C = \frac{1}{j\omega C}.$$